Trích dẫn - đồng hồ theo dõi sức khỏe

Mọi số liệu dùng trong báo cáo và schematic đều trỏ về một trang của tài liệu trong thư mục `datasheets/`. Số trang = trang PDF (trùng số in trên trang; riêng HDG: số in = PDF − 3). **ASSUME** = giả định, chưa có tài liệu.

1. Tài liệu

Khóa	Tài liệu	File
DS	Espressif ESP32-S3 Series Datasheet v2.2	`datasheets/ESP32-S3_Datasheet.pdf`
HDG	Espressif ESP32-S3 Hardware Design Guidelines	`datasheets/ESP32-S3_Hardware_Design_Guidelines.pdf`
AXP	X-Powers AXP2101 datasheet	`datasheets/LCSC_C3036461_AXP2101.pdf`
MAX	Analog Devices MAX86140/MAX86141 datasheet	`datasheets/LCSC_C5328762_MAX86141.pdf`
M30	Maxim MAX30208 datasheet	`datasheets/LCSC_C2874216_MAX30208.pdf`
BMI	Bosch BMI270 datasheet BST-BMI270-DS000-08	`datasheets/Bosch_BMI270_datasheet.pdf`
TPS	TI TPS61240/TPS61241 datasheet SLVS806D	`datasheets/LCSC_C133991_TPS61240.pdf`
TXS	TI TXS0108E datasheet SCES642L	`datasheets/LCSC_C17206_TXS0108E.pdf`
LP	TI LP5907 datasheet	`datasheets/LCSC_C92498_LP5907-1.8.pdf`
W25	Winbond W25Q128JV datasheet	`datasheets/LCSC_C97521_W25Q128JVSIQ.pdf`
SFH	OSRAM SFH 7016 datasheet v1.5	`datasheets/OSRAM_SFH7016.pdf`
VEMD	Vishay VEMD8080 datasheet rev 1.0	`datasheets/LCSC_C511402_VEMD8080.pdf`
LC6	ST USBLC6-2 datasheet	`datasheets/LCSC_C7519_USBLC6-2SC6.pdf`
TYC	USB-C receptacle TYPE-C-31-M-12 drawing	`datasheets/LCSC_C165948_TYPE-C-31-M-12.pdf`
XT	40 MHz crystal X322540MPB4SI	`datasheets/LCSC_C9010_X322540MPB4SI_40MHz.pdf`
TS	tact switch TS-1088	`datasheets/LCSC_C720477_TS-1088_tact_switch.pdf`
AO	AO3401A (P-FET, reference only)	`datasheets/LCSC_C15127_AO3401A.pdf`
RTC	NXP PCF85063A datasheet Rev. 7.3	`datasheets/PCF85063A.pdf`
ANT	Johanson 2450AT18D0100 datasheet Ver. 4.4	`datasheets/Johanson_2450AT18D0100.pdf`
FPC	HDGC 0.3K-HX-nPWB FPC connector drawing	`datasheets/HDGC_0.3K-HX-nPWB.pdf`
WS	Waveshare ESP32-S3-Touch-AMOLED-1.75C schematic (reference design)	`datasheets/Waveshare_ESP32-S3-Touch-AMOLED-1.75C_schematic.pdf`

2. Số liệu đã đối chiếu tự động (29/29 khớp trang PDF)

Số liệu	Giá trị	Tài liệu, trang	Đoạn văn bản phải có trên trang	Kiểm
ESP32-S3 light-sleep (Table 5-10)	0.24 mA	DS p.68 (ESP32-S3_Datasheet.pdf)	`Light-sleep1?\s+VDD_SPI and Wi-Fi are powered down.*?240`	OK
ESP32-S3 deep-sleep (Table 5-10)	0.007 mA	DS p.68 (ESP32-S3_Datasheet.pdf)	`RTC memory is powered up\. RTC peripherals are powered down\.\s+7`	OK
ESP32-S3 160 MHz, 2 nhân chạy (Table 5-9, Typ2)	64.1 mA	DS p.67 (ESP32-S3_Datasheet.pdf)	`160\s+Dual core running 32-bit data access instructions\s+49\.6\s+64\.`	OK

Số liệu	Giá trị	Tài liệu, trang	Đoạn văn bản phải có trên trang	Kiểm
ESP32-S3 240 MHz, 2 nhân chạy (Table 5-9, Typ2)	81.3 mA	DS p.67 (ESP32-S3_Datasheet.pdf)	`240\s+Dual core running 32-bit data access instructions\s+66\.2\s+81\.`	OK
WiFi TX đỉnh HT20 MCS7 (Table 5-7)	283 mA	DS p.66 (ESP32-S3_Datasheet.pdf)	`802\.11n, HT20, MCS7, @18\.5 dBm\s+283`	OK
WiFi RX (Table 5-7)	88 mA	DS p.66 (ESP32-S3_Datasheet.pdf)	`802\.11b/g/n, HT20\s+88`	OK
BLE TX 0 dBm đỉnh (Table 5-8)	176 mA	DS p.67 (ESP32-S3_Datasheet.pdf)	`Bluetooth LE @ 0 dBm\s+176`	OK
MAX86141 ILED, 1 LED 62 mA, 25 sps	0.185 mA	MAX p.7 (LCSC_C5328762_MAX86141.pdf)	`PPG_SR=25sps\s+185`	OK
MAX86141 ILED, 1 LED 62 mA, 100 sps	0.735 mA	MAX p.7 (LCSC_C5328762_MAX86141.pdf)	`PPG_SR = 100sps\s+735`	OK
MAX86141 IDD, 25 sps, LP_MODE	0.011 mA	MAX p.7 (LCSC_C5328762_MAX86141.pdf)	`PPG_SR = 25sps\s+11`	OK
MAX86141 IDD shutdown	0.0006 mA	MAX p.7 (LCSC_C5328762_MAX86141.pdf)	`VDD Current in Shutdown\s+TA = \+25°C\s+0\.6`	OK
MAX86141 VDD 1.7-2.0 V	1.8 V	MAX p.6 (LCSC_C5328762_MAX86141.pdf)	`1\.7\s+1\.8\s+2\.0\s+V`	OK
MAX86141 VLED 3.1-5.5 V	3.1 V	MAX p.6 (LCSC_C5328762_MAX86141.pdf)	`LED Supply Voltage\s+VLED\s+Verified during PSRR Test\s+3\.1\s+5\.5`	OK
MAX86141 SPI tối đa	4 MHz	MAX p.8 (LCSC_C5328762_MAX86141.pdf)	`SCLK Frequency\s+fSCLK\s+4\s+MHz`	OK
MAX86141 VIH	1.4 V	MAX p.8 (LCSC_C5328762_MAX86141.pdf)	`Input Voltage High\s+VIH\s+SDI, SCLK, CSB, GPIO1, GPIO2\s+1\.4`	OK
MAX86141 abs max VDD	2.2 V	MAX p.4 (LCSC_C5328762_MAX86141.pdf)	`VDD_DIG to GND_ANA[. ]+-0\.3V to \+2\.2V`	OK
SFH 7016 VF xanh lá max	2.8 V	SFH p.5 (OSRAM_SFH7016.pdf)	`Forward voltage.*?max\.\s+2\.80 V`	OK
BMI270 acc+gyro normal	0.685 mA	BMI p.11 (Bosch_BMI270_datasheet.pdf)	`A\+G Normal Mode.*?685`	OK
BMI270 acc normal (ODR max)	0.21 mA	BMI p.11 (Bosch_BMI270_datasheet.pdf)	`Aonly Normal Mode.*?210`	OK
BMI270 địa chỉ I2C	104	BMI p.128 (Bosch_BMI270_datasheet.pdf)	`0b1101000 \(0x68\)`	OK
TPS61240 Iq	0.03 mA	TPS p.5 (LCSC_C133991_TPS61240.pdf)	`Operating quiescent current\s+IOUT = 0 mA, Power save mode\s+30\s+40`	OK
TPS61240 CIN	2.2 uF	TPS p.12 (LCSC_C133991_TPS61240.pdf)	`2\.2 µF, X5R, 6\.3 V, 0402`	OK
LP5907 Iq	0.012 mA	LP p.6 (LCSC_C92498_LP5907-1.8.pdf)	`VEN = 1\.2 V, IOUT = 0 mA\s+12\s+25`	OK
MAX30208 standby	0.0005 mA	M30 p.1 (LCSC_C2874216_MAX30208.pdf)	`0\.5µA Standby Current`	OK

Số liệu	Giá trị	Tài liệu, trang	Đoạn văn bản phải có trên trang	Kiểm
AXP2101 dòng pin khi tắt (chỉ RTCLDO)	0.04 mA	AXP p.11 (LCSC_C3036461_AXP2101.pdf)	`IBAT\s+Battery discharge current\s+40`	OK
AXP2101 kéo lên I2C	2.2 kΩ	AXP p.9 (LCSC_C3036461_AXP2101.pdf)	`needs a 2\.2kΩ Pull High`	OK
AXP2101 tụ VSYS	22 uF	AXP p.8 (LCSC_C3036461_AXP2101.pdf)	`Connect two\s+22uF capacitors`	OK
TXS0108E pull-up nội	4 kΩ	TXS p.18 (LCSC_C17206_TXS0108E.pdf)	`40kΩ when the output is driving low.*?4kΩ when the output is driving h`	OK
TXS0108E VCCA ≤ VCCB	0	TXS p.20 (LCSC_C17206_TXS0108E.pdf)	`ensure that VCCA ≤ VCCB at all times`	OK

3. Trích dẫn theo linh kiện - Health watch - main board

Ref	Giá trị	Tài liệu (khóa §/bảng/trang)
J101	TYPE-C-31-M-12	TYC; HDG §1.3.13
R101	5.1k	USB Type-C spec (Rd = 5.1 kΩ)
R102	5.1k	USB Type-C spec
U101	USBLC6-2SC6	LC6 p.1; HDG §1.3.13
R103	22	HDG §1.3.13; WS p.1 (R13)
R104	22	HDG §1.3.13; WS p.1 (R14)
TP101	TP	-
U102	AXP2101	AXP p.1 (features), Table 4-1 p.7-9 (chân), Fig 7-1 p.57 (mạch ứng dụng); WS p.1
C101	10uF 10V	AXP Fig 7-1 p.57
C102	10uF 10V	AXP Fig 7-1 p.57; WS p.1 (CP2)
C103	22uF 6.3V	AXP Table 4-1 p.8 (pin 34)
C104	22uF 6.3V	AXP Table 4-1 p.8 (pin 34)
C105	1uF	AXP Table 4-1 p.8 (pin 33)
C106	1uF	AXP Fig 7-1 p.57
C107	1uF	AXP Fig 7-1 p.57
C108	2.2uF	WS p.1 (CP13); AXP Fig 7-1 p.57
C109	1uF	AXP Fig 7-1 p.57
C110	1uF	AXP Fig 7-1 p.57
L101	1uH 2.5A	AXP Fig 7-1 p.57, §6.6 p.22 (3 MHz)
C111	22uF 6.3V	AXP Fig 7-1 p.57
L102	1uH 2.5A	WS p.1 (LP2, RP1, RP2); AXP Fig 6-1 p.17, Fig 7-1 p.57
R105	0R 0805	WS p.1 (RP2); AXP p.12
R126	0R 0805	WS p.1 (RP1); AXP Fig 6-1 p.17
RT101	NTC 10k B3950	AXP Table 4-1 p.8 (pin 31 TS); WS p.1 (RP4)
R106	4.7k	AXP Table 4-1 p.8 (pin 38)
R107	2.2k	AXP Table 4-1 p.9 (pin 39 "needs a 2.2kΩ pull high")
R108	2.2k	AXP Table 4-1 p.9 (pin 40)

Ref	Giá trị	Tài liệu (khóa §/bảng/trang)
R109	510	WS p.1 (RP6)
C112	1nF	WS p.1 (CP14)
SW101	TS-1088 4x3mm	AXP Table 4-1 p.8 (pin 30 PWRON, kéo lên 100k nội)
R110	1k	AXP Fig 7-1 p.57 (LED từ VSYS về CHGLED)
D101	Red 0603	AXP Table 4-1 p.7 (pin 1)
J102	LiPo 1S ~400mAh	-
R111	0.05R 1% 0805	yêu cầu đo dòng (user)
TP102	TP	-
TP103	TP	-
R112	0.1R 1% 0603	yêu cầu đo dòng (user)
TP104	TP	-
TP105	TP	-
TP106	TP	-
TP107	TP	-
U103	TPS61240DRVT	TPS p.1, p.3 (chân), §10.2.2 p.12 (CIN 2.2 µF, COUT 4.7 µF, L 1 µH); MAX p.6 (VLED 3.1-5.5 V); SFH p.5 (VF green max 2.80 V), p.8 (đồ thị VF)
L103	1uH 1A	TPS §10.2.2 p.12 (L = 1 µH)
C113	2.2uF	TPS §10.2.2 p.12
C114	4.7uF	TPS §10.2.2 p.12
R113	1M	DS §3 Table 3-1 p.32 (GPIO3 strapping); TPS p.5 (dòng EN)
R114	0.1R 1% 0603	yêu cầu đo dòng (user)
TP108	TP	-
TP109	TP	-
U104	ESP32-S3R8	DS p.2 (features), Table 1-1 (R8 = 8 MB PSRAM), Table 2-1 (chân)
C115	10uF 10V	HDG §1.3.2 p.8
C116	1uF	HDG Fig.1 p.5
C117	100nF	HDG Fig.1 p.5
L104	2.0nH	HDG §1.3.2 p.8; WS p.1 (IND1 2.0nH)
C118	1uF	HDG Fig.1 p.5
C119	10nF	HDG Fig.1 p.5
C120	100nF	HDG §1.3.2
C121	100nF	HDG §1.3.2
C122	100nF	HDG §1.3.2 p.7
C123	1uF	HDG §1.3.2 p.7
R115	10k	HDG §1.3.3 Table 2 p.8; WS p.1 (R8)
C124	1uF	HDG §1.3.3 p.8
SW102	TS-1088 4x3mm	HDG §1.3.3
R116	10k	DS §3 Table 3-1 p.32; HDG §1.3.9
SW103	TS-1088 4x3mm	DS Table 3-3 p.33; HDG §1.3.9
Y101	40MHz 10ppm CL15pF	HDG §1.3.5 p.10; XT

Ref	Giá trị	Tài liệu (khóa §/bảng/trang)
L105	24nH	HDG §1.3.5 p.10; WS p.1 (R10 24nH)
C125	22pF C0G	HDG §1.3.5 p.11
C126	22pF C0G	HDG §1.3.5 p.11
U105	W25Q128JVSIQ	W25 p.1, §3 (chân SOIC-8 208 mil); DS §3.2 Table 3-4 p.34 (VDD_SPI theo GPIO45); WS p.1 (U2)
C127	100nF	W25
C128	1.5pF C0G	HDG §1.3.6 Table 6 p.13
L106	2.4nH	HDG §1.3.6 Table 6 p.13
C129	1.5pF C0G	HDG §1.3.6 p.12-13
C130	DNP	HDG §1.3.6 p.12
R117	0R	HDG §1.3.6 p.12
C131	DNP	HDG §1.3.6 p.12
AE101	2450AT18D0100E	HDG §1.3.6; ANT p.1-2
R118	499	HDG §1.3.7 p.13
J103	UART0 1x03 2.54	yêu cầu user
J104	0.3K-HX-31PWB	WS p.1 (J3, bảng GPIO); FPC p.1-2
C132	10uF 10V	ASSUME (thực hành chung)
C133	100nF	-
U106	BMI270	BMI Table 22 p.135 (chân, cách nối I2C), §5 p.128 (địa chỉ 0x68)
C134	100nF	BMI §7 (sơ đồ nối) - ASSUME giá trị 100 nF
C135	100nF	BMI §7
U107	PCF85063ATT/AJ	RTC Table 3 (chân), §7.2.2.4 (COF)
Y102	32.768kHz CL7pF	RTC p.1 (CL = 7 pF hoặc 12.5 pF)
C136	100nF	-
R119	10k	-
Q101	AO3400A	AO3400A datasheet (ASSUME, chưa tải)
R120	100	-
R121	100k	DS §3 Table 3-1 p.32, Table 3-3 p.33 (GPIO46)
M101	Coin motor 1027 3V	-
D102	BAT54WS	-
Q102	AO3400A	-
R122	100	-
R123	100k	DS §3 Table 3-1 p.32, Table 3-4 p.34 (GPIO45)
BZ101	CMT-8504-100-SMT-TR	DigiKey CMT-8504-100-SMT-TR (datasheet chưa tải được)
D103	BAT54WS	-
SW104	TS-1088 4x3mm	DS §2.3 p.23-24 (RTC GPIO)
R124	10k	-
C137	100nF	-
R125	1k	-
D104	Green 0603	yêu cầu user

Ref	Giá trị	Tài liệu (khóa §/bảng/trang)
TP110	TP	-
TP111	TP	-
TP112	TP	-
TP113	TP	-
TP114	TP	-
J105	FH12-16S-0.5SH	Hirose FH12 (footprint KiCad)

3. Trích dẫn theo linh kiện - Health watch - sensor board

Ref	Giá trị	Tài liệu (khóa §/bảng/trang)
J201	FH12-16S-0.5SH	-
C201	10uF 10V	-
C202	10uF 10V	MAX p.12 (VLED: bypass 10 µF to PGND)
C203	100nF	ASSUME
U201	LP5907MFX-1.8	LP p.1 (noise, PSRR); MAX p.6 (VDD 1.7-2.0 V), p.7 (shutdown 0.6 µA)
C204	1uF	LP p.6 (điều kiện đo CIN = COUT = 1 µF)
C205	1uF	LP p.7 (COUT, ESR)
R201	0.1R 1% 0603	yêu cầu đo dòng (user)
TP201	TP	-
TP202	TP	-
TP203	TP	-
U202	MAX86141ENP+T	MAX p.1, p.6 (VLED, VDD), p.8 (SPI 4 MHz, VIH 1.4 V), p.11 (bump map), p.12 (chân)
C206	1uF	MAX p.12 (VREF)
C207	100nF	MAX p.12 (VDD_ANA)
C208	10uF 10V	MAX p.12 (VDD_ANA)
C209	100nF	MAX p.12 (VDD_DIG); sơ đồ khối MAX p.40
D201	SFH 7016	SFH p.3-5 (bước sóng, VF), p.11-12 (xung), p.13-14 (chân, pad); MAX p.12 (nối LED)
D202	VEMD8080	VEMD p.1, p.4 (chân); MAX p.12 (PD1_IN)
D203	VEMD8080	VEMD p.4; MAX p.12 (PD2_IN, chỉ MAX86141)
R205	0R	MAX p.12 (PD_GND), p.88 (shorting via / 0 Ω, Note 3)
U203	TXS0108EPWR	TXS p.1, p.4 (chân), p.6 (OE: VIH, Δt/Δv ≤ 10 ns/V), §7.3.5 p.18 (pull-up 4k/40k nội), §8.3 p.20 (VCCA ≤ VCCB); MAX p.8 (mức logic), DS Table 5-4 p.65 (VIH = 0.75·VDD)
C211	100nF	TXS §8.4
C212	100nF	TXS §8.4
U204	MAX30208CLB+	M30 p.1, p.2 (reflow, sai số), p.6 (chân), p.7 (lắp đặt), Table 3 p.10 (địa chỉ)
C213	100nF	M30 p.6 (VDD)
R203	10k	M30 Table 2-3 p.10
R204	10k	M30 Table 2-3 p.10